

Skin Lesion Diagnosis System Using Multimodal Deep Learning

Final Project Report

1. Introduction

Skin cancer is one of the most common cancers worldwide. This project presents a multimodal deep learning system that classifies dermoscopic images into seven classes using EfficientNet-B4 together with clinical metadata (age, sex, and lesion location).

2. Problem Statement

The project addresses visual similarity between lesions, severe class imbalance, and the need to combine image information with clinical context for more reliable diagnosis.

3. Dataset

The HAM10000 dataset was used. The data were split into 6,009 training images, 2,003 validation images, and 2,003 testing images. Clinical metadata included age, sex, and lesion location.

4. Data Preprocessing

Missing ages were imputed using the median. Sex and lesion location missing values were mapped to 'Unknown'. Age was normalized, sex encoded, and lesion location one-hot encoded into an 18-dimensional clinical vector.

5. Class Balancing

Minority classes were oversampled until each class contained 4,023 training images. WeightedRandomSampler was also used during training.

6. Data Augmentation

Training images were augmented using random flips, rotation, resized crop, affine transformations, color jitter, random erasing, and ImageNet normalization.

7. Model Architecture

The model combines an EfficientNet-B4 image branch with an MLP (18→64→32) clinical branch. Features are concatenated and passed through fully connected layers to classify seven skin lesion classes.

8. Training Experiments

Two configurations were evaluated. The baseline (Adam + Cross Entropy + ReduceLROnPlateau) achieved the best validation Macro F1 (0.8254), outperforming AdamW + Focal Loss + Cosine Annealing.

9. Results

Final test performance: Accuracy 86.9%, Macro F1 0.786, Weighted F1 0.870. The highest F1 was achieved for NV (0.94), while MEL remained the most challenging class.

10. Explainability

Grad-CAM heatmaps highlight lesion regions responsible for predictions, improving model transparency.

11. Deployment

The solution was deployed with FastAPI. Test-Time Augmentation averages predictions from original, horizontal-flip, and vertical-flip images. Outputs include diagnosis, confidence scores, risk tier, and an independent ABCDE checklist.

12. Tools and Technologies

Python, PyTorch, EfficientNet-B4, Pandas, NumPy, Scikit-learn, OpenCV, FastAPI, Google Colab, CUDA.

13. Future Work

Future improvements include lesion segmentation, mobile deployment, calibrated confidence, medical chatbot, and additional classifiers.

14. Conclusion

The multimodal approach effectively combines image and clinical information, achieving strong performance while providing explainable predictions suitable for preliminary decision support.